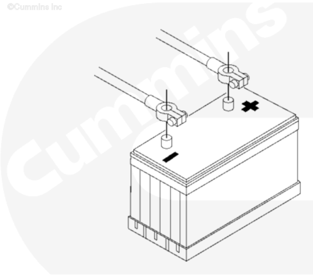
## Trainings

## Preparatory Steps

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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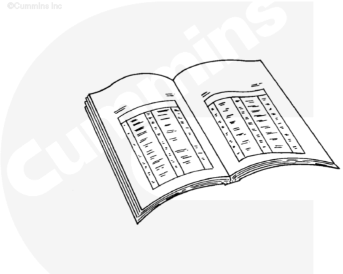
Disconnect the batteries.

**⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**



ck800wa

**⚠ WARNING ⚠**

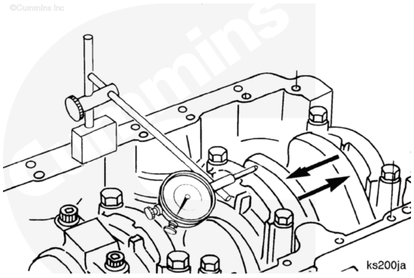
**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

- Drain the lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 (/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- If equipped, remove the block stiffener plate. Refer to Procedure 001-089

(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html).

## Initial Check

**Note : The dimensions of the thrust bearing and crankshaft journal determine end clearance.**

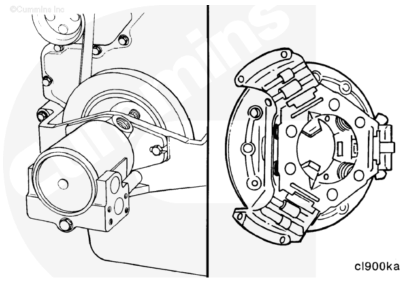



Measure the crankshaft end clearance using dial indicator, Part Number 3824564 and magnetic base, Part Number 3377399.

| Crankshaft End clearance | | |
|---|---|---|
| mm | | in |
| 0.102 | MIN | 0.004 |
| 0.432 | MAX | 0.017 |

If the crankshaft end clearance is **not** within specification, make sure to inspect the crankshaft and thrust bearing surfaces for damage.

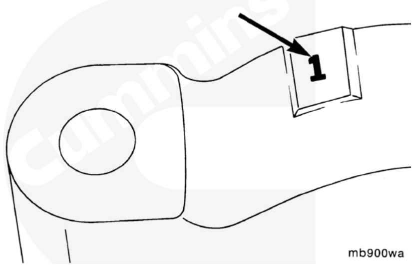
A common cause for increased crankshaft end clearance and thrust bearing damage is increased end-loading of the engine. The increased end-loading can be the result of driven units at the front or rear of the engine being:




- Incorrectly installed
- Incorrectly adjusted
- Incorrectly matched to the engine and exceeding the thrust load limits.

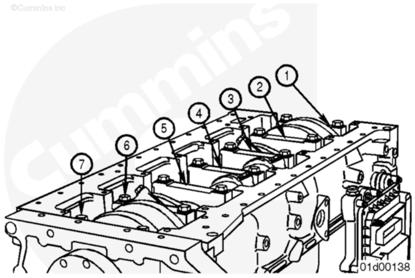
## Remove

Before removing the main bearing caps, make certain that the caps are clearly marked for their location on the lubricating oil cooler side of the main bearing cap and cylinder block.




The number one cap is at the front of the engine.

**Note : When replacing bearings in chassis: For four cylinder engines, replace number 2 through 4 while the number 1 and number 5 caps support the crankshaft. After replacing number 2 through number 4, replace number 1 and number 5. For six cylinder engines, replace number 2 through 6 while the number 1 and number 7 caps support the crankshaft. After replacing number 2 through number 6, replace number 1 and number 7.**




Loosen the main bearing capscrews completely, but do **not** remove.

**⚠ CAUTION ⚠**

**Do not pry on the main bearing caps to free them from the cylinder block. Damage to the main bearing caps and cylinder block can result.**

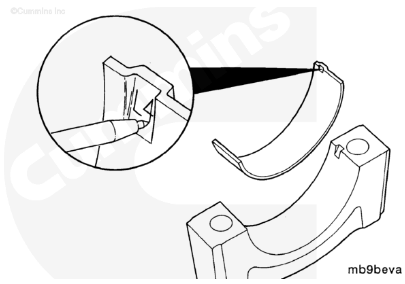

Use two of the main bearing cap bolts to “wiggle” the main bearing cap loose, being careful **not** to damage the bolt threads.

Remove the main bearing cap.

Mark the main bearings for position and number as they are removed.

Use an awl to mark the bearing's position in the tang area.

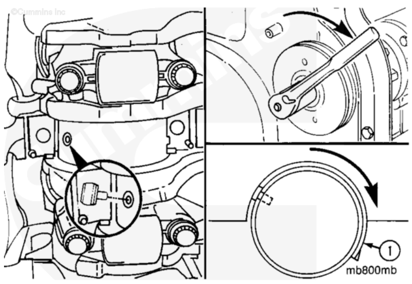
**Note : Marking the bearing's position is for future identification or possible failure analysis.**




Follow this step to remove the upper main bearings, except for number 1 front main bearing.

To remove the upper main bearing, install the main bearing replacer, Part Number 3823818, in the oil hole of the crankshaft main bearing journal.

Using barring tool, part number 3824591, rotate the crankshaft so that the replacer contacts the upper main bearing on the side opposite the tang.

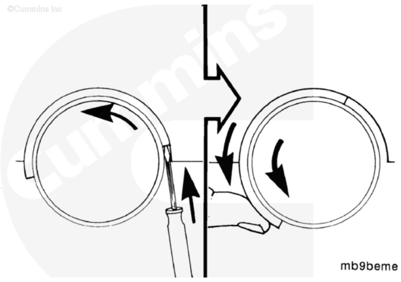


Continue to rotate the crankshaft in the direction that will remove the tang side (1) of the upper main bearing first.

Remove the bearing.

**⚠ CAUTION ⚠**

**Use care so the screwdriver does not damage the crankshaft or cylinder block.**



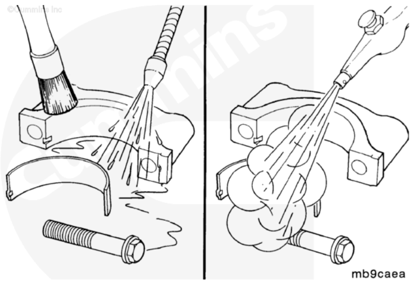

**Note : The front main bearing, number 1, does not have a hole in the journal, so the tool can not be used to replace the bearing.**

Using a flat blade screwdriver, gently bump the end of the bearing to loosen it from the cylinder block. Then, use finger pressure against the main bearing shell and rotate the crankshaft to roll the main bearing out.

## Clean and Inspect for Reuse

**⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**




**⚠ WARNING ⚠**

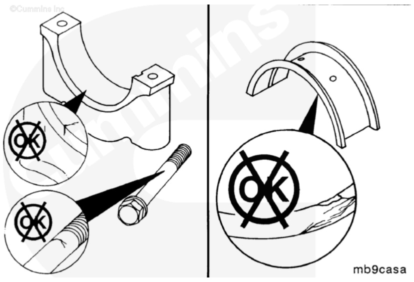
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Steam-clean or use hot, soapy water to clean the main bearing caps.

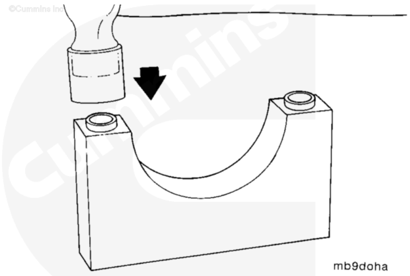
Dry with compressed air.

Inspect all main bearing caps, capscrews and thrust bearings for deep scoring, overheating, etc.

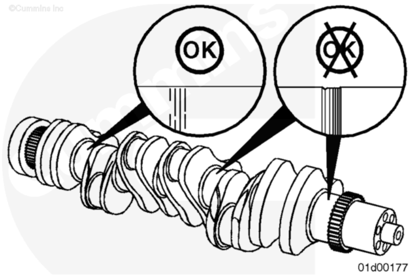
Replace any damaged components. If the main bearing cap is damaged, the block **must** be replaced.




Check the main bearing caps to make sure the ring dowels are installed.




Check the crankshaft main bearing journals for damage or excessive wear. Minor scratches are acceptable.




If crankshaft end clearance measured during the initial check was found to be out of specification, make sure to visually check the crankshaft thrust surface for excessive wear or damage. Minor scratches are acceptable.

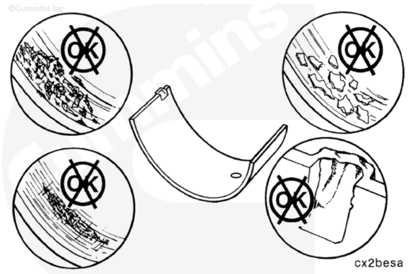
Four cylinder engines — The number 4 main bearing journal.

Six cylinder engines — The number 6 main bearing journal.

If damage is found, the crankshaft will need to be removed. Refer to Procedure 001-016 (/qs3/pubsys2/xml/en/procedures/40/40-001-016-tr.html).

Also visually check the thrust bearing surfaces for excessive wear. Replace the thrust bearing(s) if excessive wear is found

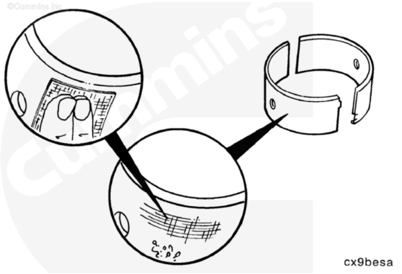
Inspect the bearings for damage.




Replace any bearings with the following damage:

- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches.

Inspect the bearing shell seating surfaces for nicks or burrs. If nicks or burrs can **not** be removed with an abrasive pad, Part Number 3823258 or equivalent, the bearings **must** be replaced.

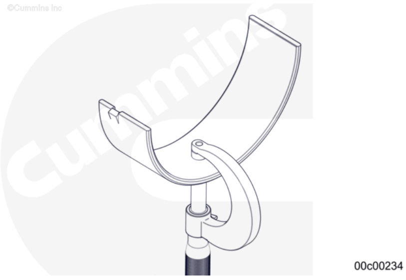



**Note : If bearings are damaged they must be replaced as a set.**

**Note : For more detailed information on bearing damage, see "Analysis and Prevention of Bearing Failures" Bulletin 3810387.**

## Measure

Measure the main bearing shell thickness with an outside micrometer that has a ball tip.



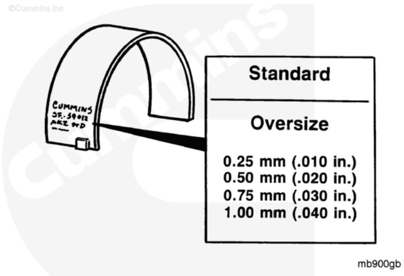

| Main Bearing Dimensions | | | |
|---|---|---|---|
| | mm | | in |
| Standard | 2.446 | MIN | 0.0963 |
| | 2.464 | MAX | 0.0970 |
| 0.25 mm Oversize | 2.571 | MIN | 0.1012 |
| | 2.589 | MAX | 0.1019 |
| 0.50 mm Oversize | 2.696 | MIN | 0.1061 |
| | 2.714 | MAX | 0.1068 |
| 0.75 mm Oversize | 2.821 | MIN | 0.1111 |
| | 2.839 | MAX | 0.1118 |
| 1.00 mm Oversize | 2.946 | MIN | 0.1160 |
| | 2.964 | MAX | 0.1167 |

Discard a bearing shell if its thickness is below the minimum specifications.

**Note : Main bearings are identified with a part number and size stamped on the back side.**

If replacing the bearings, determine the size of the removed main bearings and obtain a set of the same size.

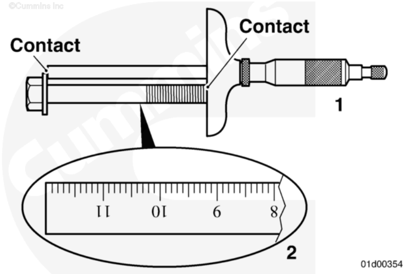
Oversize service main bearings are available for use with crankshafts that have been machined undersize. See the appropriate parts catalog.




Main Bearing Capscrew Reuse Measurement

**CAUTION**

**This step must be completed on B4.5 RGT engines. Failure to check the main bearing capscrew against reuse guidelines can result in severe engine damage.**




To check if a main bearing capscrew can be reused, the length **must** be measured by performing the following:

For each main bearing capscrew that has been removed, measure the length from underneath the head of the capscrew to the tip of the capscrew, as illustrated, using one of two methods.

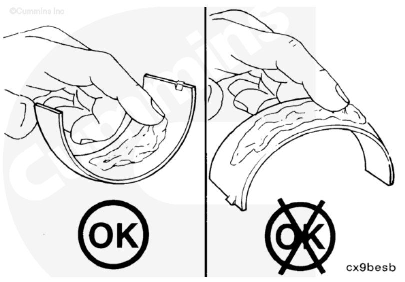
1. A depth micrometer (preferred method for accuracy)
2. A machinist's rule.

If the measurement is above the maximum specification, the main bearing capscrew **must** be replaced.

| Main Bearing Underhead Capscrew Length | | |
|---|---|---|
| mm | | in |
| 120.00 | MAX | 4.724 |

# Install

Upper Main Bearing




**Note : Make sure the main bearing being installed is the same size as the main bearing removed, if the crankshaft has not been machined. The size is engraved on the back of the main bearing.**

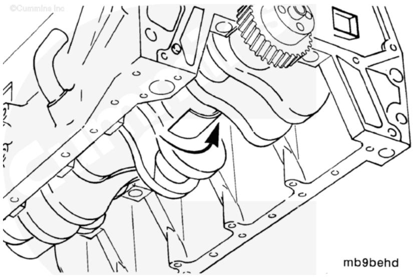
**Note : Install the upper main bearing cap after each upper main bearing is installed to keep the main**

**bearing in place while the other upper main bearings are installed.**

**⚠ CAUTION ⚠**

**Do not lubricate the side that is against the cylinder block.**

Apply assembly lube, Part Number 3163086, to the upper main bearings.




**Note : The crankshaft thrust bearing must be installed in the: Four cylinder engines - The number 4 main bearing position. Six cylinder engines - The number 6 main bearing position.**

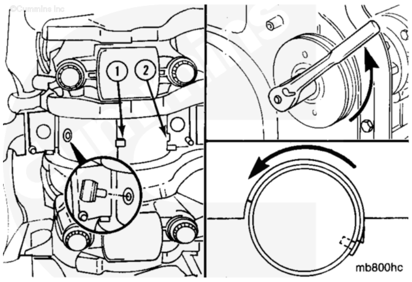
**Note : Some engines use a thrust bearing for the upper and lower main bearing (360 degree), while other engines only have a thrust bearing for the upper main bearing (180 degree). Always replace like for like.**

**Note : The upper and lower main bearing shells are not interchangeable. The backs of the main bearings are marked with the proper orientation.**

Install the upper main bearings.

Insert the side of the main bearing opposite the tang first in between the crankshaft journal and block. Install the bearing as far as possible by hand.

When installing the thrust bearing, it may be necessary to push the crankshaft to the front or rear of the cylinder block.




**⚠ CAUTION ⚠**

**Make sure the pin does not slide under the bearing.**

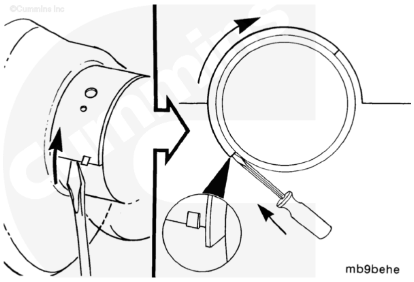
Follow this step to finish installing the upper main bearings, except for number 1 front main bearing.

Using the main bearing replacer, Part Number 3823818, finish installing the main bearing. Rotate the crankshaft, using the barring tool, part number 3824591.

Make sure the tang (1) on the main bearing is located in the notch (2) of the cylinder block. Finish pushing the main bearing into position.

**⚠ CAUTION ⚠**

**Use care so the screwdriver does not damage the crankshaft or cylinder block.**




**Note : The front main, number 1, does not have a hole in the journal so the pin can not be used to replace the bearing.**

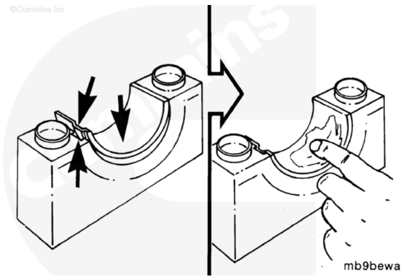
Install the number 1 main bearing.

Insert the side of the main bearing opposite the tang first and install as far as possible by hand.

Use a flat blade screwdriver to push the main bearing into position while rotating the crankshaft.

Make sure the tang on the main bearing is located in the notch of the cylinder block.

Lower Main Bearings




**⚠ CAUTION ⚠**

**Do not lubricate the back side of the bearing that contacts the main bearing cap.**

Making sure that the backside of the bearings are clean and free of debris, install the lower main bearings into the main bearing caps.

Make sure to align the tangs of the bearings with tangs on the main bearing caps.

**Note : Some engines use a thrust bearing for the upper and lower main bearing (360 degree), while other engines only have a thrust bearing for the upper main bearing (180 degree). Always replace like for like.**

If equipped, install the lower crankshaft thrust bearing.

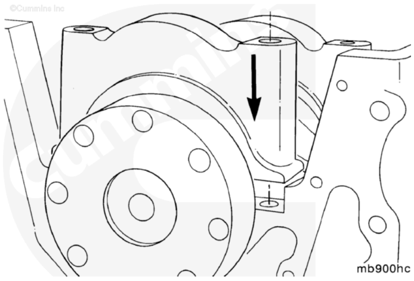
Four cylinder engines - The number 4 main bearing position.

Six cylinder engines - The number 6 main bearing position.

Apply a coat of assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

**⚠ CAUTION ⚠**

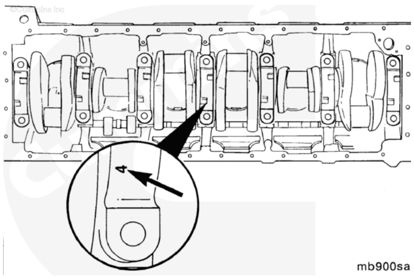
**Make sure the caps are correctly installed in the same position as removed, with the number towards the oil cooler side of the engine.**




Make sure the main bear cap surfaces between the main bearing cap and block are clean and free of debris.

Install the main bearing cap into position, lining the main bearing cap dowel rings with the cylinder block.

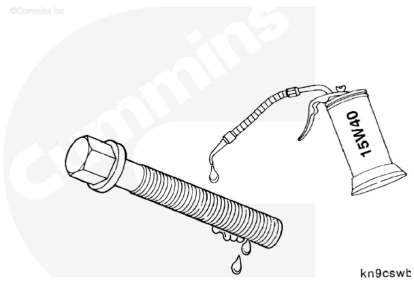
The main bearing caps are/were numbered during the removal process for their location. Number 1 starts with the front of the block.



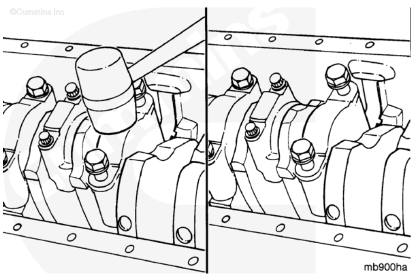

**Note : The caps must be installed so the numbers on the caps match the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap must be on the same side.**

Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.

Lubricate the main bearing capscrew threads and underside of the head with clean engine oil.




Gently tap the main bearing cap into position with a plastic or rubber mallet.




When seated, install the main bearing capscrews and tighten.

**Torque Value:** 50 n•m [ 37 ft-lb ]

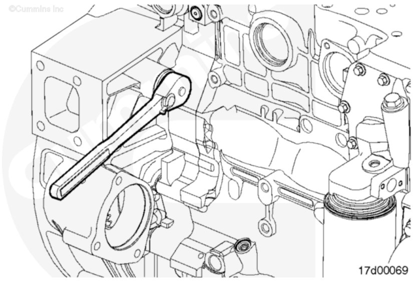
Do **not** tighten to the final torque value at this time. Final torque should be applied after all main bearing caps are installed.

Using the barring tool, part number 3824591, the crankshaft **must** rotate freely after installing the main bearing caps.

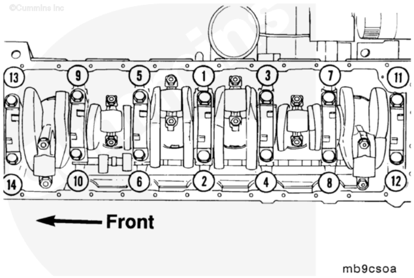
While applying final torque to the main bearing capscrews, frequently check that the crankshaft rotates freely.

If the crankshaft does **not** rotate freely:

1. Check if the crankshaft is contacting one of the connecting rods
2. Check if the correct main bearing caps where installed correctly
3. Check if the main bearing cap ring dowels or mounting surfaces were damaged during installation
4. Check if the correct main bearings where installed.




**Note : The sequence to the right is for a six cylinder engine. For a four cylinder engine, use the same sequence for the 5 main bearing caps.**




Tighten the main bearing capscrews evenly and in sequence.

B3.9, B4.5 and B5.9 Main Bearing Capscrews

**Torque Value:**

1. 60 n•m [ 44 ft-lb ]
2. 90 n•m [ 66 ft-lb ]
3. Rotate 90 degrees.

**⚠ CAUTION ⚠**

**For B4.5 RGT engines there is a different torque procedure for new and previously installed main bearing capscrews. Failure to use the correct torque process can result in engine damage.**

B4.5 RGT Main Bearing Capscrews

**Torque Value:**
Previously Installed Main Bearing Capscrews

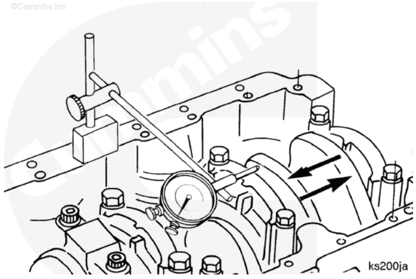
1. 60 n•m [ 44 ft-lb ]
2. 80 n•m [ 59 ft-lb ]
3. Rotate 90 degrees.

**Torque Value:**
New Main Bearing Capscrews

1. 120 n•m [ 89 ft-lb ]
2. Loosen completely
3. 60 n•m [ 44 ft-lb ]
4. 85 n•m [ 63 ft-lb ]
5. Rotate 120 degrees.

**Note : The dimensions of the thrust bearing and crankshaft journal determine end clearance.**

Measure the crankshaft end clearance using dial indicator, Part Number 3824564 and magnetic base, Part Number 3377399.




| Crankshaft End clearance | | |
|---|---|---|
| mm | | in |
| 0.102 | MIN | 0.004 |
| 0.432 | MAX | 0.017 |

If the crankshaft end clearance is **not** within specification:

1. If the crankshaft end clearance is below specification, check if there are any obstructions limiting the crankshaft's travel (lubricating oil pump, connecting rod, etc.)
2. If the crankshaft end clearance is above specification, inspect the crankshaft thrust bearing surface. Also check if the correct thrust bearing(s) were installed.

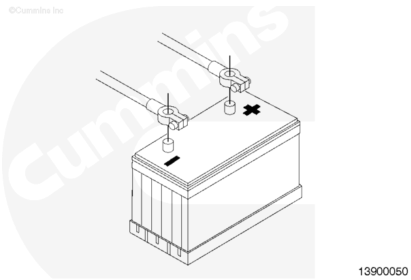
**Note : Oversize thrust bearings are available if the end clearance is not within specifications. Oversize thrust bearings of 0.25 to 0.51 mm [0.010 to 0.020 in] are available.**

## Finishing Steps


- If equipped, install the block stiffener plate. Refer to Procedure 001-089 (/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 (/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Fill the lubricating oil pan. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html).

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**




Connect the batteries.

Operate the engine at idle for 5 to 10 minutes. Check for loose parts, leaks and proper oil pressure.

**Last Modified: 06-Oct-2016**